

MA_vax: comparing the four fits

Massachusetts 2025–26 influenza season. Four calibrations of the same model against the same hospital-admission data:

- **7-age** vs **1-age**: the age-structured model (7 groups) vs the single-group model built by population-weighting its inputs.
- **m(t)** vs **no m(t)**: with and without the fitted time-varying transmission multiplier (18 log-increments on 14-day knots).

All intervals are the median and 2.5–97.5 percentile across each fit's posterior draws (one simulation per accepted parameter set, **baseline** vaccination scenario).

1. Headline comparison

Quantity	7-age, m(t)	7-age, no m(t)	1-age, m(t)	1-age, no m(t)
Posterior draws	627	429	495	352
Infection attack rate	6.7% [5.0% – 10.4%]	36.9% [32.9% – 40.8%]	6.4% [2.3% – 34.6%]	38.5% [32.5% – 44.2%]
Cumulative hospitalizations (simulated)	6,070 [5,569 – 6,549]	5,622 [4,943 – 6,389]	6,060 [5,110 – 7,075]	6,225 [5,359 – 7,190]
Cumulative hospitalizations (observed)	6,157	6,157	6,134	6,134
Difference vs observed	-1.4%	-8.7%	-1.2%	+1.5%
Infections per hospitalization	78 [58 – 121]	458 [394 – 532]	74 [28 – 402]	434 [359 – 499]
IHR, population-weighted	2.098% [1.355% – 2.799%]	0.331% [0.279% – 0.393%]	1.438% [0.266% – 3.865%]	0.246% [0.215% – 0.297%]
IHR, infection-weighted	1.440% [0.931% – 1.930%]	0.244% [0.210% – 0.285%]	1.438% [0.266% – 3.865%]	0.246% [0.215% – 0.297%]
IHR_scale, population-weighted	0.83 [0.53 – 1.13]	0.15 [0.13 – 0.17]	0.68 [0.13 – 1.82]	0.12 [0.10 – 0.14]
Peak daily hospitalizations (simulated)	165 [142 – 190]	81 [67 – 97]	135 [96 – 189]	81 [60 – 106]
Peak daily hospitalizations (observed)	183	183	183	183

Quantity	7-age, m(t)	7-age, no m(t)	1-age, m(t)	1-age, no m(t)
beta_baseline	0.0303 [0.0251 – 0.0374]	0.0368 [0.0312 – 0.0411]	0.0324 [0.0235 – 0.0511]	0.0403 [0.0342 – 0.0444]
humidity_impact	0.60 [0.14 – 0.95]	0.21 [0.01 – 0.54]	0.35 [0.02 – 0.89]	0.12 [0.01 – 0.48]
seed_scale_E	1.57 [0.37 – 3.72]	2.03 [0.65 – 7.34]	1.02 [0.15 – 7.87]	2.23 [0.64 – 8.36]
phi (NB dispersion)	55 [15 – 306]	3 [2 – 5]	10 [2 – 39]	2 [1 – 4]
Peak R_eff	2.09	1.71	1.97	1.59
Peak R_eff date	2025-12-08	2025-10-09	2025-12-08	2025-10-09
R_eff, season end	0.45	0.43	0.54	0.45
R_v, season end (no infection-acquired immunity)	0.48	0.69	0.60	0.74
R_eff / R_v, season end (share of R left after depletion)	0.93	0.62	0.90	0.61
Season-mean effective beta (per contact per day)	0.0317	0.0392	0.0359	0.0415

Two IHR averages appear above and they answer different questions. **Population-weighted** IHR is $\Sigma_a (N_a/N) \cdot IHR_a$ — the risk faced by a randomly chosen *resident*. It depends only on the fitted parameters and the age distribution, not on the epidemic, and in an age-structured model it is dominated by 65+ (1.70 of the 7-age m(t) fit's 2.10 percentage points come from that one group). **Infection-weighted** IHR is $\Sigma_a (infections_a/infections) \cdot IHR_a$ — the risk carried by a randomly chosen *infection*, i.e. the number that multiplied by total infections gives total hospitalizations.

They differ whenever infections are not distributed like the population. In the 7-age fits, 65+ takes 17.5% of the population but only ~10.6% of infections (low contact rates), while 5-12 takes 8.7% of the population and ~10.4% of infections — so the infection-weighted average sits well below the population-weighted one (1.44% vs 2.10%). In the single-age model there is no age structure, so the two are identical by construction. **Across models, compare the infection-weighted row:** it is 1.440% (7-age) against 1.438% (1-age), which is exactly the agreement you would expect from two fits matching the same hospitalizations with the same attack rate. The population-weighted row differing (2.10% vs 1.44%) is a statement about demography and where the fit put the age-specific scaling, not a disagreement about severity.

2. Infection attack rate

Ever-infected fraction at the end of the season, $(N - S - SV) / N$.

Age group	7-age, m(t)	7-age, no m(t)	1-age, m(t)	1-age, no m(t)
0-0	4.8% [3.6% – 7.4%]	26.9% [23.7% – 30.0%]	—	—
1-4	5.3% [3.9% – 8.1%]	30.0% [26.7% – 33.4%]	—	—
5-12	8.1% [6.0% – 12.3%]	44.3% [40.1% – 48.3%]	—	—
13-17	8.3% [6.2% – 12.6%]	45.1% [40.8% – 49.1%]	—	—
18-49	7.6% [5.7% – 11.7%]	41.0% [36.6% – 45.2%]	—	—
50-64	6.5% [4.9% – 10.1%]	35.7% [31.7% – 39.7%]	—	—
65+	4.1% [3.0% – 6.4%]	24.0% [21.0% – 27.1%]	—	—
All	6.7% [5.0% – 10.4%]	36.9% [32.9% – 40.8%]	6.4% [2.3% – 34.6%]	38.5% [32.5% – 44.2%]

3. Hospitalizations: fitted vs observed

Cumulative new hospital admissions over the season (**I_to_H** + **IV_to_H**), against the calibration target.

Age group	7-age, m(t) (simulated)	7-age, no m(t) (simulated)	1-age, m(t) (simulated)	1-age, no m(t) (simulated)	Observed
0-0	36 [22 – 48]	42 [27 – 58]	—	—	41
1-4	153 [104 – 201]	167 [120 – 217]	—	—	165
5-12	156 [111 – 204]	162 [112 – 208]	—	—	161
13-17	95 [63 – 123]	93 [64 – 125]	—	—	90
18-49	746 [582 – 895]	771 [653 – 933]	—	—	727
50-64	753 [595 – 911]	740 [589 – 916]	—	—	750
65+	4,125 [3,700 – 4,555]	3,630 [3,076 – 4,335]	—	—	4,224
All	6,070 [5,569 – 6,549]	5,622 [4,943 – 6,389]	6,060 [5,110 – 7,075]	6,225 [5,359 – 7,190]	6,157

Age group	7-age, m(t) (simulated)	7-age, no m(t) (simulated)	1-age, m(t) (simulated)	1-age, no m(t) (simulated)	Observed
% difference	-1.4%	-8.7%	-1.2%	+1.5%	

Difference from observed, by age group

Age group	7-age, m(t)	7-age, no m(t)
0-0	-13.0%	+1.5%
1-4	-6.8%	+1.7%
5-12	-2.6%	+0.6%
13-17	+5.3%	+2.4%
18-49	+2.7%	+6.1%
50-64	+0.4%	-1.4%
65+	-2.3%	-14.1%
All	-1.4%	-8.7%

The single-age fits are calibrated to the population total only (2 targets), the 7-age fits to the 7 age series plus an end-of-season by-age scalar (8 targets), so the by-age columns are a fit check for the former and a fit result for the latter.

4. IHR scaling

IHR_scale multiplies both hospitalization-risk parameters (**I_to_H_prop** and **IV_to_H_prop**) for that age group; the fitted IHR is that multiplier times the config's baseline value.

Fitted **IHR_scale**

Age group	7-age, m(t)	7-age, no m(t)	1-age, m(t)	1-age, no m(t)	ratio m(t) / no m(t)
0-0	1.54 [0.85 – 1.96]	0.32 [0.20 – 0.45]	—	—	4.8×
1-4	1.56 [0.86 – 1.97]	0.29 [0.21 – 0.39]	—	—	5.4×
5-12	1.18 [0.69 – 1.73]	0.22 [0.15 – 0.30]	—	—	5.3×
13-17	1.01 [0.55 – 1.62]	0.19 [0.13 – 0.25]	—	—	5.4×
18-49	0.59 [0.36 – 0.85]	0.11 [0.10 – 0.14]	—	—	5.2×

Age group	7-age, m(t)	7-age, no m(t)	1-age, m(t)	1-age, no m(t)	ratio m(t) / no m(t)
50-64	0.78 [0.46 – 1.15]	0.14 [0.11 – 0.18]	—	—	5.5×
65+	1.07 [0.69 – 1.42]	0.16 [0.13 – 0.19]	—	—	6.7×
All (pop-weighted)	0.83 [0.53 – 1.13]	0.15 [0.13 – 0.17]	0.68 [0.13 – 1.82]	0.12 [0.10 – 0.14]	5.5× / 5.8×

Fitted IHR (unvaccinated arm, **I_to_H_prop** after scaling)

Age group	Config baseline	7-age, m(t)	7-age, no m(t)	1-age, m(t)	1-age, no m(t)
0-0	0.697%	1.077% [0.594% – 1.366%]	0.226% [0.142% – 0.312%]	—	—
1-4	0.697%	1.086% [0.599% – 1.375%]	0.201% [0.147% – 0.269%]	—	—
5-12	0.274%	0.322% [0.188% – 0.473%]	0.061% [0.042% – 0.081%]	—	—
13-17	0.274%	0.278% [0.151% – 0.444%]	0.051% [0.036% – 0.069%]	—	—
18-49	0.561%	0.331% [0.201% – 0.478%]	0.063% [0.057% – 0.079%]	—	—
50-64	1.060%	0.823% [0.489% – 1.217%]	0.149% [0.116% – 0.191%]	—	—
65+	9.091%	9.712% [6.316% – 12.951%]	1.459% [1.197% – 1.766%]	—	—
All (pop-weighted)	2.118%	2.098% [1.355% – 2.799%]	0.331% [0.279% – 0.393%]	1.438% [0.266% – 3.865%]	0.246% [0.215% – 0.297%]

Fitted IHR (vaccinated arm, **IV_to_H_prop** after scaling)

Age group	7-age, m(t)	7-age, no m(t)	1-age, m(t)	1-age, no m(t)
0-0	0.982% [0.542% – 1.246%]	0.206% [0.130% – 0.285%]	—	—
1-4	0.990% [0.547% – 1.254%]	0.183% [0.134% – 0.246%]	—	—
5-12	0.294% [0.172% – 0.432%]	0.056% [0.039% – 0.074%]	—	—
13-17	0.254% [0.138% – 0.405%]	0.047% [0.032% – 0.063%]	—	—
18-49	0.301% [0.183% – 0.435%]	0.058% [0.052% – 0.072%]	—	—
50-64	0.750% [0.446% – 1.109%]	0.136% [0.105% – 0.174%]	—	—
65+	6.701% [4.358% – 8.936%]	1.007% [0.826% – 1.219%]	—	—

5. What actually separates the fits

The m(t) and no-m(t) fits imply the same hospitalizations from ~5.5× different epidemics.

Hospitalizations are infections × IHR, and hospital-admission data alone identifies only that product. The two families split it very differently:

- **Without m(t)**, susceptible depletion is the only mechanism in the model that can bend the epidemic curve over — R_v (transmission before infection-acquired immunity) stays flat near 1.25–1.35 all season. To reproduce the observed peak-and-decline the fit must burn through a large share of the population, so it lands on a ~37–38% attack rate and a correspondingly low IHR.
- **With m(t)**, the multiplier can push transmission down directly (R_v falls from ~1.5 in December to ~0.8 in January), so the curve turns over with almost no depletion — a ~6.5% attack rate, and an IHR ~5.5× higher to compensate.

The IHR_scale ratio between the two families tracks the attack-rate ratio almost exactly, age group by age group (§4), which is the signature of a near-degenerate product rather than two genuinely different fits.

This is a substantive difference, not a technicality. It changes the implied infection burden of the season by a factor of ~5.5 and the implied severity by the same factor in the other direction. A ~6.5% season attack rate is low for influenza and a ~37% one is on the high side; breaking the degeneracy needs an infection-side constraint (serology, ILI, wastewater, or an explicit prior on attack rate or IHR) — the hospitalization series cannot do it.

The no-m(t) fits fit visibly worse, and the dispersion parameter says so. ϕ_i (NB2 dispersion — smaller means more overdispersion, i.e. more of the data written off as observation noise) drops from 55 [15 – 306] to 3 [2 – 5] in the 7-age pair and from 10 to 2 in the single-age pair. Without m(t) the model

cannot track the shape of the curve, so the likelihood buys agreement with slack instead. The 7-age no- $m(t)$ fit also misses the total by -8.7% and 65+ by -14.1% (§3), while the $m(t)$ fit is within -1.4% / -2.3%. That is not by itself grounds to prefer $m(t)$: it adds 18 free parameters under a random-walk prior, so a fair comparison needs WAIC or a marginal likelihood, not just the residuals.

The single-age $m(t)$ fit is the least well identified of the four. Its attack rate spans 2.3% – 34.6% across the posterior and its `IHR_scale` spans 0.13 – 1.82 — the whole degeneracy described above, visible *within* one posterior rather than between fits. The 7-age $m(t)$ fit is much tighter (5.0% – 10.4%), presumably because the by-age targets constrain the infection distribution. Treat single-age $m(t)$ posterior means with care.

Age structure matters much less than $m(t)$. The 7-age and 1-age fits agree closely on attack rate, hospitalizations and R_{eff} within each $m(t)$ family; the population-weighted collapse loses the contact assortativity (which shows up as a slightly higher R for the same effective beta — see `effective_beta_MA_vax.png`) but not much else at this level of aggregation.

Caveats. Attack rate is $(N - S - SV) / N$ at the final day, so it counts anyone who ever left the susceptible arms by infection. Per-draw trajectories are paired to their parameter sets by reversing the `default_rng(42)` permutation the run scripts use; `compute_effective_beta.py` asserts that pairing on every run. The simulation outputs on disk are rewritten whenever the corresponding `run_simulations_*` script or counterfactual notebook is re-run, so regenerate this report after any such run.